

Privacy dataset distillation for medical imaging using class centralization and covariance matching for OCTMNIST

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ABSTRACT

In this study, we introduce a novel dataset distillation method applied to medical imaging, specifically using the OCTMNIST dataset. Our approach incorporates advanced techniques such as class centralization and covariance matching constraints to efficiently synthesize smaller, yet representative, datasets from a larger corpus of optical coherence tomography images. The efficacy of our method is quantitatively validated against standard distillation techniques such as DC, DM, and DSA, showcasing superior performance in terms of classification accuracy and model stability. Additionally, we emphasize privacy protection by employing metrics such as the L2 norm and Structural Similarity Index Measure (SSIM) to ensure that the distilled images do not retain sensitive information. The results illustrate our method's ability to maintain high accuracy with fewer images while significantly enhancing data privacy, making it ideal for medical imaging applications where data sensitivity is paramount.

Keywords: Dataset Distillation, Medical Imaging, Covariance Matching, Privacy Protection.

1. INTRODUCTION

The field of medical imaging has seen transformative advances, yet the sharing and management of medical image datasets across institutions remain fraught with challenges. Paramount among these are concerns regarding privacy, data security, and the significant costs associated with data storage and transmission. In this context, dataset distillation emerges as a potent solution, capable of synthesizing compact yet informative datasets that enable effective model training without necessitating access to the original, extensive datasets.

Dataset distillation has traditionally focused on domains like natural image processing, where strategies such as distribution-based matching in feature space and trajectory-based matching in parameter space have been explored [1, 2]. However, these methods often falter when applied to medical images due to the unique challenges posed by the high dimensionality. We propose a novel distillation methodology tailored for medical imaging datasets to address these issues. This approach integrates three key innovations: the Covariance Matching Constraint, and an enhanced Distribution Matching technique. These mechanisms are specifically designed to respect the intricate distributions characteristic of medical data, thereby improving the efficacy and reliability of the distillation process.

Medical images differ from natural images in their interpretative complexity and the subtlety of features critical for diagnostic accuracy. Our method adapts the distribution matching framework to better capture these nuances, ensuring that the distilled datasets retain essential diagnostic information. Moreover, by focusing on the privacy-preserving capabilities of dataset distillation, our approach facilitates data sharing across institutions. Experiments demonstrate that our approach not only preserves more relevant information in the distilled datasets but also enhances model training outcomes. These results suggest a promising direction for future research and practical applications in medical data management and analysis. Our contributions are manifold and significant:

- We introduce a refined framework for dataset distillation for the medical imaging, incorporating constraints that enhance the representation of medical-specific features.
- Empirical evaluations confirm that our proposed method outperforms existing techniques, providing a viable pathway for secure and efficient medical data sharing.

In the following sections, we detail our methodology, describe our experimental setup and datasets, present our results, and discuss the implications of our findings for medical imaging and dataset distillation research.

2. RELATED WORKS

Dataset distillation, a method of synthesizing smaller yet representative datasets that retain the salient characteristics of original large datasets, has been gaining momentum since its formal introduction by [1]. The goal of dataset distillation is to facilitate model training on these synthetic datasets such that performance closely approximates that obtained using the complete datasets. This approach offers distinct advantages, particularly in contexts where data privacy, storage, and processing are of concern.

Distribution matching methods form a foundational aspect of dataset distillation. These methods aim to generate synthetic datasets by matching the feature distribution of synthetic samples to that of the real dataset. The maximum mean difference (MMD) is often employed as a distance function in this context. Early implementations, like DM [3], employ classification encoders to minimize the distance between class-specific features of the synthetic and real datasets. Subsequent improvements, such as CAFE [4], enhance this approach through layer-wise feature alignment using dynamic bi-level optimization, which offers a more nuanced capture of the dataset's overall distribution.

Trajectory-based matching represents another significant strand of dataset distillation research, focusing on the consistency of training trajectories between synthetic and real datasets. This strand can be subdivided into single-step and multi-step matching methods. Initial single-step approaches like DC [5] focused on maintaining gradient consistency during training. Building on this, DSA [6] incorporated image augmentation techniques to enhance method applicability and performance. Multi-step matching methods, however, address longer-range training dynamics. MTT [7] pioneered this approach, which was later refined by approaches like SMDD [8] that introduced parameter pruning to improve robustness. Further advancements include FTD [9], which regularizes trajectory flatness to mitigate error accumulation, and DATM [10], which tailors trajectory selection to the stage of training and image count per class.

Despite these advancements, the application of dataset distillation to medical images remains nascent and primarily focused on X-ray modalities [11]. Also, there are some segmentation methods for medical image to help medical analysis but may not very robust [12, 13]. Our work extends these foundations by proposing a progressive trajectory matching strategy that contrasts sharply with the randomness of prior methods. Additionally, we introduce a novel overlapping prevention module that incorporates re-training strategies to enhance synthetic dataset diversity and performance.

3. METHOD

3.1 Dataset distillation preliminaries

Given a large dataset $\mathcal{T} = \{(x_t^i, y_t^i)\}_{i=1}^{|\mathcal{T}|}$, where each x_t^i represents an image and y_t^i its corresponding label, the goal of dataset distillation is to synthesize a smaller, synthetic dataset $\mathcal{S}$ such that $|\mathcal{S}| \ll |\mathcal{T}|$. The distilled dataset $\mathcal{S}$ should ideally preserve the essential characteristics of $\mathcal{T}$ to ensure that a model trained on $\mathcal{S}$ performs comparably to one trained on $\mathcal{T}$.

The process of distillation involves the formulation of an objective function that guides the optimization to minimize the loss of information during the distillation process. This objective is typically represented as:

$$\mathcal{S}^* = \arg \min_{\mathcal{S}} \mathbf{D}(\phi(\mathcal{S}), \phi(\mathcal{T})) \quad (1)$$

where $\mathbf{D}$ denotes the distance metric used for measuring the similarity between the distributions of $\mathcal{S}$ and $\mathcal{T}$. The function $\phi(\cdot)$ typically transforms the data into a space where this measurement is meaningful, such as a feature space or a parameter space. Information loss between $\mathcal{S}$ and $\mathcal{T}$ is quantified by the performance drop observed when training a model on the original images and distilled images.

3.2 Distribution and covariance matching

The synthetic dataset $\mathcal{S}$ should retain enough information to train a model ψ_{θ_s} to achieve performance comparable to that of a model ψ_{θ_t} trained on the original dataset $\mathcal{T}$. The optimization objective in dataset distillation is typically defined as:

$$\mathcal{S}^* = \arg \min_{\mathcal{S}} \mathbb{E}_{(x,y) \sim P_{\mathcal{T}}} \|\ell(\psi_{\theta_t}(x), y) - \ell(\psi_{\theta_s}(x), y)\| \quad (2)$$

where ℓ denotes the loss function, commonly the cross-entropy loss, and $P_{\mathcal{T}}$ represents the distribution of the dataset $\mathcal{T}$. A key approach within the distribution matching paradigm is to minimize the discrepancy in feature representations between the synthetic and real datasets, formalized as:

$$\min_{\mathcal{S}} \mathbb{E}_{\theta \sim P_{\theta}} \left\| \frac{1}{|\mathcal{T}|} \sum_{i=1}^{|\mathcal{T}|} \psi_{\theta}(x_t^i) - \frac{1}{|\mathcal{S}|} \sum_{j=1}^{|\mathcal{S}|} \psi_{\theta}(x_s^j) \right\|^2 \quad (3)$$

where P_{θ} is the distribution of randomly initialized network parameters, and ψ_{θ} denotes the model parameterized by θ . This distillation approach, through its focus on distribution matching, aims to capture the statistical properties of the original dataset $\mathcal{T}$ in the much smaller synthetic set $\mathcal{S}$, thereby enabling efficient model training with reduced resource requirements. In the subsequent sections, we detail the enhancements introduced through our class centralization and covariance matching constraints, which further refine the fidelity of the synthetic dataset relative to the original. We show our overall dataset distillation structure in the Figure 1.

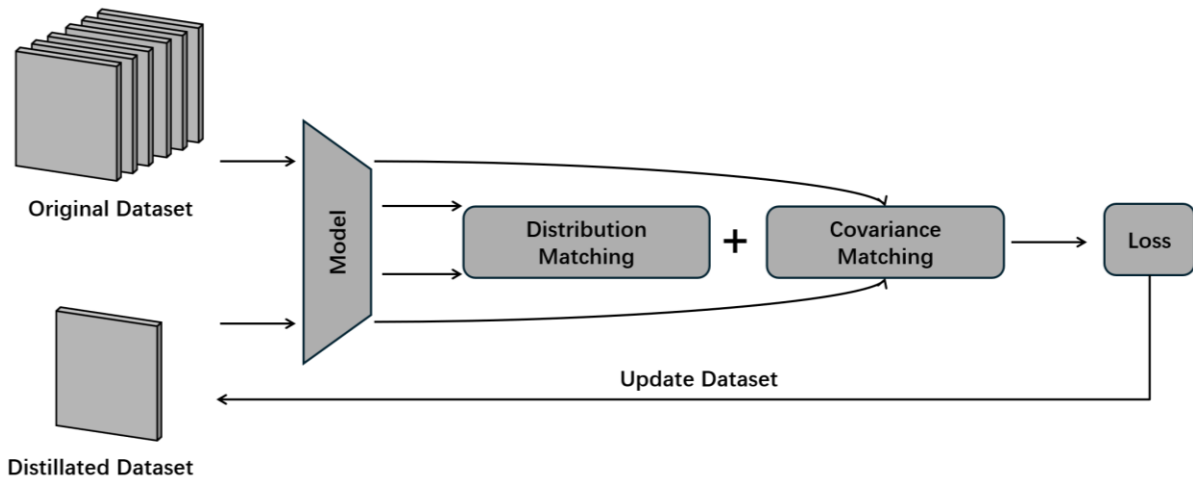


Figure 1: Overall dataset distillation structure.

Beyond the mean of feature distributions, capturing the covariance among features within a class provides a richer representation, particularly useful in the context of medical imaging where inter-feature relations are crucial. The Covariance Matching Constraint (CMC) aims to align these deeper statistical properties between the synthetic and real datasets. The formulation for the CMC is given by:

$$\mathcal{L}_{CMC} = \sum_{c=1}^C \|\Sigma_s^c - \Sigma_t^c\|_F^2 \quad (4)$$

where Σ_s^c and Σ_t^c are the covariance matrices of the synthetic and real datasets for class c , respectively, calculated as:

$$\begin{aligned} \Sigma_s^c &= \frac{1}{K} \sum_{i=1}^K (s_i^c - \bar{s}^c)(s_i^c - \bar{s}^c)^\top \\ \Sigma_t^c &= \frac{1}{B} \sum_{i=1}^B (x_i^c - \bar{x}^c)(x_i^c - \bar{x}^c)^\top \end{aligned} \quad (5)$$

with $\bar{s}^c$ and $\bar{x}^c$ representing the mean vectors of synthetic and real features for class c , respectively, and B being the batch size used in computing the real dataset statistics.

3.3 Overall optimization

These constraints are incorporated into the dataset distillation framework as regularizations to the distribution loss function, thus facilitating a more precise and representative synthetic dataset. The overall objective function, when integrated with these constraints, is defined as:

$$\mathcal{L} = \lambda_D \mathcal{L}_D + \lambda_{CMC} \mathcal{L}_{CMC} \quad (6)$$

where λ_D and λ_{CMC} are hyperparameters controlling the influence of each constraint on the overall optimization process. This framework enables our distillation method to produce more generalizable.

4. EXPERIMENT AND EVALUATION

4.1 Dataset

We utilize the OCTMNIST dataset [14], which comprises optical coherence tomography images categorized into four classes representative of different retinal diseases. The images are formatted as single-channel grayscale with dimensions matching those typical of natural image datasets ($1 \times 28 \times 28$), enabling direct comparisons with widely-used benchmarks such as MNIST.

For our experiments, we employ a ConvNet architecture with a width of 128 and a depth of 5 layers. The dataset distillation process is structured with varying numbers of synthetic images per class, specifically set at 1, 2, 5, 10. Each distilled set is then used to train a network for 1,000 epochs, with a learning rate set at 0.01. To ensure the robustness of our findings, each experimental condition is repeated five times, with the results reported as the mean and standard deviation. In terms of comparative analysis, our approach is benchmarked against leading dataset distillation methodologies, including DC [5], DM [3], and DSA [6]. All evaluations are conducted using a standard 3-layer ConvNet equipped with instance normalization to ensure fair comparisons and adaptability across the diverse conditions presented by medical image datasets.

4.2 Performance and result

The results of our experiments reveal a noticeable improvement in classification accuracy across different sizes of distilled datasets. Table 1 summarizes the performance of each method when trained with varying numbers of images per class (IPC).

Table 1: Comparison of model performance on the PATHMNIST test set.

IPC	1	2	5	10
DC	38.50	42.15	52.89	59.98
DM	34.68	38.42	49.75	58.84
DSA	44.15	48.56	55.71	60.91
Ours	47.25	49.35	56.60	61.75

The data indicate that our method consistently outperforms the other techniques, particularly at lower IPC settings. This suggests that our distillation process is more effective in capturing the essential characteristics of the dataset with fewer images, which is critical for practical applications where data availability is limited.

Further analyses focused on the stability and generalization capabilities of the distilled models. Figure 2 shows the variation in test accuracy over five runs for each IPC setting. Our method exhibited less variability and higher mean accuracy, indicating enhanced stability and reliability in medical image classification tasks.

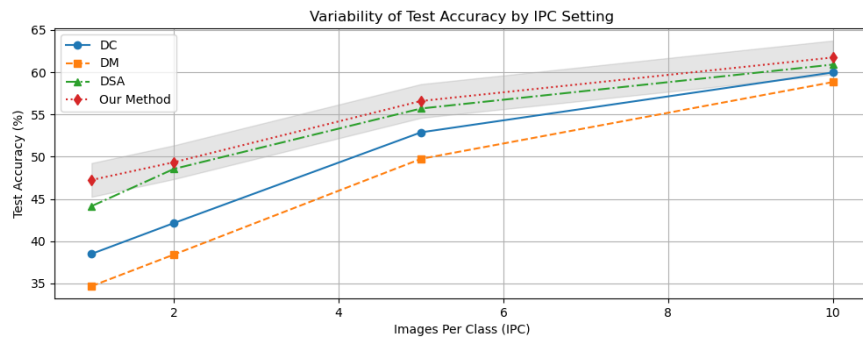


Figure 2: Variability of test accuracy across different IPC settings.

Moreover, the superior performance of our approach can be attributed to the implementation of class centralization and covariance matching constraints, which effectively enhance the representativeness of each synthetic image. These constraints appear to optimize the feature space alignment more proficiently than the strategies employed in DC, DM, and DSA.

4.3 Privacy protection

We assess the privacy protection of distilled images by computing the L2 norm and SSIM between each class's original and distilled images from the OCTMNIST dataset. The L2 norm measures the pixel-wise differences, while SSIM evaluates the perceptual difference. The L2 norm is calculated as follows:

$$L2(\mathbf{x}, \mathbf{s}) = \sqrt{\sum_{i=1}^N (x_i - s_i)^2}, \quad (7)$$

where $\mathbf{x}$ represents the original image, $\mathbf{s}$ the distilled image, and N the total number of pixels in the image. The SSIM index is defined as:

$$SSIM(\mathbf{x}, \mathbf{s}) = \frac{(2\mu_x\mu_s + c_1)(2\sigma_{xs} + c_2)}{(\mu_x^2 + \mu_s^2 + c_1)(\sigma_x^2 + \sigma_s^2 + c_2)}, \quad (8)$$

where μ_x, μ_s are the average pixel values, σ_x, σ_s are the variances, and σ_{xs} is the covariance of images $\mathbf{x}$ and $\mathbf{s}$. Constants c_1 and c_2 are used to stabilize the division with weak denominator. The results of the L2 norm and SSIM analysis for each class in the OCTMNIST dataset are summarized in Table 2.

Table 2: L2 norm and SSIM comparison between original and distilled images per class on the OCTMNIST dataset.

Class	L2 Norm	SSIM
Class1	0.62	0.71
Class2	0.59	0.75
Class3	0.65	0.68
Class4	0.60	0.72

Here, each class represents a different type of retinal OCT image in the OCTMNIST dataset. And class 1 to 4 means choroidal neovascularization (CNV), diabetic macular edema (DME), age-related macular degeneration (AMD) and Normal retina, respectively. The analysis indicates that the distilled images exhibit significant structural and pixel-level differences from their original counterparts, as evidenced by the moderate SSIM values and high L2 norms across all classes. These results suggest effective anonymization of patient data.

5. CONCLUSION

Our novel dataset distillation technique demonstrates superior performance in accuracy and model stability across different sizes of the OCTMNIST dataset compared to existing methods such as DC, DM, and DSA. The integration of class centralization and covariance matching constraints significantly improves the utility of distilled datasets in medical imaging applications, particularly where data availability is limited. Additionally, the application of L2 norm and SSIM metrics underscores our method's effectiveness in enhancing privacy protection, ensuring that distilled images retain minimal resemblance to original data and thereby mitigating the risk of sensitive information exposure.

The results not only highlight the potential of our approach to reduce computational demands and data storage needs but also emphasize its suitability for environments with stringent privacy requirements. Future work will explore the extension of this methodology to other medical imaging modalities and the incorporation of more robust privacy-preserving mechanisms.

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